Anomaly Detection in Industrial Networks: A Comparison of Random Forest and Isolation Forest

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***Abstract*—Industrial networks require reliable anomaly detection to protect critical infrastructure from disruptions and cyberattacks. This study compares two machine learning methods for flow-level anomaly detection in industrial network traffic: Random Forest, a supervised method, and Isolation Forest, an unsupervised method. Both methods are evaluated under three training approaches; centralized, local and federated. The experiments are conducted on the Westermo dataset, collected from a physical industrial network testbed, and validated on the Edge-IIoT dataset to assess generalizability. Results show that Random Forest consistently outperforms Isolation Forest across all training strategies and both datasets. Federated training underperforms relative to centralized and local approaches, primarily due to non-IID data distributions and limitations in aggregation across distributed nodes. Isolation Forest demonstrates value in detecting volumetric anomalies but struggles with subtle attack types, suggesting its best role is as part of a hybrid detection system. The findings indicate that the performance advantage of supervised methods holds across datasets with different network characteristics and attack compositions.**

***Index Terms*—Anomaly Detection, Federated Learning, Industrial Networks, Isolation Forest, Network Intrusion Detection, Random Forest**

I. Introduction

Within industrial and critical societal systems, such as energy, transportation, water supply and manufacturing, the requirement for stable and secure network communication is essential. Industrial networks differ from traditional IT networks in that they are often geographically distributed, operate over long periods of time, and require a higher degree of availability and reliability. In these environments, disruptions can lead to serious consequences, as physical processes may be directly affected when data is lost or compromised.

Industrial networking solutions are often deployed in distributed systems, where security monitoring and anomaly detection become particularly challenging. Different parts of the network may have separate functions and traffic patterns, and collecting all data at a central location is often impractical. This creates a need for efficient and practical methods for detecting anomalies, meaning abnormal behavior in the network. The need for anomaly detection in industrial networks has been widely discussed due to the constraints and security challenges of such environments [1], [2].

This study is carried out in collaboration with Westermo and utilizes their publicly available network traffic dataset [3], which was collected from a physical testbed resembling an industrial communication system. To address the challenge of detecting anomalous behavior in industrial network traffic, this study investigates machine learning-based anomaly detection using flow-level traffic analysis. Specifically, the study compares Random Forest, a supervised method, with Isolation Forest, an unsupervised method, under three training approaches: centralized, local and federated. Federated learning is of particular interest because it enables collaborative model training across distributed nodes without requiring the exchange of raw data, which aligns with the constraints of industrial deployments.

II. Background

A. Industrial networks

Industrial networks provide the communication infrastructure between devices in critical sectors, including energy production, transportation, manufacturing and water treatment [4]. These sectors carry a responsibility to ensure safety for operations such as communication, production continuity and equipment integrity; since failure could lead to serious consequences for society and the environment. Industries therefore operate under strict requirements for availability, determinism and reliability, with systems designed to support real-time monitoring, control and physical processes [4].

Industrial networks differ from conventional IT environments in several ways. Traffic patterns are typically predictable and repetitive, as devices exchange periodic control messages and sensor data [4]. Systems are often geographically distributed, resource-constrained or located in harsh environments, which demands careful design choices to maintain effective communication.

With the introduction of the Industrial Internet of Things (IIoT), industrial networks have grown more complex; they now contain more devices and connection types while also handling internal and external access, monitoring and maintenance [4]. This expansion has increased the attack surface. In such interconnected environments, sending all data to a single location is not always viable due to security, bandwidth and latency constraints. As networks grow in size and distribution, centralized anomaly detection becomes increasingly difficult to maintain, which has motivated the exploration of federated learning as a way to train models locally and collaborate without sharing raw network data.

B. Anomaly detection

Anomaly detection identifies patterns that deviate from normal behavior. Such deviations are not always malicious; they can also result from misconfigurations [5], [6]. This suits industrial traffic, which is repetitive and structured, but anomaly-based methods face challenges such as higher false positive rates and issues like masking and swamping [6]. Network intrusion detection systems (IDS) are divided into signature-based and anomaly-based detection [5], [6]. Signature-based detection relies on known attack patterns but cannot detect new types of attacks [6], a limitation particularly relevant in industrial systems.

Machine learning has become a common approach for anomaly detection due to its ability to model complex network patterns [6]. These methods can be divided into supervised and unsupervised learning. Supervised approaches require labeled data and perform well when attack types resemble those seen during training [5]. Unsupervised approaches instead learn patterns of normal behavior and can potentially detect previously unseen anomalies, but often at the cost of increased false alarms [6]. This distinction between supervised and unsupervised methods motivates the comparison between Random Forest and Isolation Forest in this study.

1) Supervised Learning: Random Forest: Random Forest (RF) is a supervised learning method that constructs multiple decision trees using bootstrap sampling and random feature selection at each split [7]. The final prediction is determined by majority vote across all trees, which improves generalization and overall performance.

RF has proven effective for anomaly detection and intrusion detection: it handles high-dimensional data well, maintains accuracy in the presence of noise, and models complex feature interactions with minimal parameter tuning [7]. It also provides a feature importance measure based on mean decrease in impurity. As a supervised method, RF requires labeled training data and may fail to generalize to attack types not represented in the training set.

2) Unsupervised Learning: Isolation Forest: Isolation Forest (IF) operates without labeled classifications, which gives it an inherent advantage for detecting data points not seen during training. It works by isolating anomalies through random partitioning of the feature space using randomly selected attributes and split values [8], [9]. Two assumptions underlie its operation: first, that anomalies constitute a minority of the data; second, that anomalies have attribute values that differ from normal instances [8]. As a result, anomalies require fewer splits to isolate in a tree structure, leading to shorter average path lengths.

This allows IF to detect previously unseen anomalies without labeled attack data [9]. IF is also computationally efficient, has low memory requirements and scales well using subsampling [8]. Since it flags deviations from learned patterns rather than classifying attack types, it may produce higher false positive rates.

When evaluating IF performance, two phenomena must be considered. Swamping refers to normal points being incorrectly classified as anomalies. Masking occurs when a sufficient number of anomalies form their own cluster and are consequently classified as normal [9].

C. Federated learning

Industrial networks are typically distributed across multiple locations, making centralized model training costly in terms of latency, bandwidth and privacy [10], [11]. Federated Learning (FL) addresses this by enabling collaborative training without sharing raw data [10]. Each node trains a local model and transmits only model updates to a central server, which aggregates the results and distributes the updated model back to each edge device [11]. This suits distributed anomaly detection where traffic patterns vary between nodes, though FL introduces trade-offs between communication overhead, model performance and computational limitations at the edge [10].

III. Related Work

Machine learning has been widely applied to network intrusion detection as an alternative to rule-based and signature-based systems, which struggle to adapt to evolving threats. Among the available methods, Random Forest has consistently performed well across a range of datasets and evaluation setups [12]. Perez et al. [13] evaluated Random Forest and Support Vector Machines (SVM) for attack detection in a gas pipeline SCADA system and found that Random Forest achieved higher precision and F1-score, confirming its effectiveness in industrial control system environments.

Dehlaghi-Ghadim et al. [14] compared several machine learning models for anomaly detection on the Westermo network traffic dataset, including Logistic Regression, SVM, Neural Networks, K-Nearest Neighbors, Decision Trees and Random Forest. Their results showed that Random Forest was among the best-performing classifiers for both binary and multiclass classification. The study also evaluated federated learning, where the federated Random Forest model outperformed its centralized counterpart, demonstrating that federated training can preserve data privacy while improving detection performance. Only supervised methods were evaluated, which limits the findings to anomaly types present in the labeled training data.

Isolation Forest has received increasing attention as an unsupervised alternative for network intrusion detection. Elsaid et al. [15] proposed an optimized Isolation Forest-based intrusion detection system for IIoT networks, using a modified Harris Hawks Optimization algorithm for feature selection to reduce dimensionality and improve detection. Evaluated on the NSL-KDD, UNSW-NB15 and CICIDS-2018 datasets, their system achieved improved accuracy, recall and F1-score over the standard Isolation Forest, reaching up to 99% accuracy on CICIDS-2018 dataset. The study shows that Isolation Forest can be adapted for industrial environments, though it operated in a centralized setting and did not evaluate federated training.

Federated learning has also been applied to intrusion detection in distributed environments. Mothukuri et al. [16]

evaluated a federated anomaly detection system combining GRU-based neural networks with a Random Forest ensembler on a Modbus-based dataset, achieving 99.5% accuracy while preserving data privacy. Li et al. [17] proposed a novel approach combining federated learning with Isolation Forest for anomaly detection in IoT network traffic. While their evaluation did not compare against supervised federated methods such as Random Forest, it demonstrated that a federated Isolation Forest can achieve detection performance close to a centralized model on industrial network data, including the Edge-IIoTset dataset.

IV. Research Questions

Prior work has evaluated Random Forest in federated settings [14] and Isolation Forest in centralized settings [15], but no study has compared both under the same training conditions or benchmarked federated Isolation Forest against supervised alternatives [17]. This study addresses these gaps by investigating the following research questions:

1. How does the performance of Random Forest and Isolation Forest vary across centralized, local and federated training?
2. How do Random Forest and Isolation Forest differ in which anomaly types they detect, and what are the trade-offs in precision and false alarm rate?
3. How well do the findings generalize when evaluated on a second dataset with different network characteristics and attack types?

V. Method

A. Datasets

1) Westermo Dataset: The Westermo dataset [3] was collected from a physical network testbed designed to resemble an industrial communication system. The testbed consisted of twelve devices, including Westermo routers and endpoint devices running an industrial process simulator. Network traffic was captured simultaneously at three switch nodes (Right, Bottom and Left) while seven types of network events were introduced in 16 randomized batches over approximately 90 minutes.

Six of the seven event types are considered anomalies: IP misconfiguration (BAD-MISCONF), IP duplication (BAD-MISCONF-DUPLICATION), man-in-the-middle (BAD-MITM), single-device port scan (BAD-PORTSCAN1), multi-device port scan (BAD-PORTSCAN2), and SSH brute-force (BAD-SSH). The seventh event, legitimate SSH login (Good-SSH), is relabeled as normal traffic. The raw PCAP files were processed into network flow data using the ICS-FlowGenerator tool, resulting in three CSV files with a total of 48,657 flows across 60 features. The dataset is imbalanced, with approximately 75% normal traffic. Each node naturally contains different traffic distributions due to its position in the network topology, making the dataset suitable for evaluating federated and local training approaches.

2) Edge-IIoT Dataset: The Edge-IIoT dataset [18] was generated from an IoT/IIoT testbed with multiple sensor types and communication protocols. Features were extracted from network traffic using the Zeek and TShark protocol analyzers, yielding 61 features selected for high correlation from an initial set of 1,176 attributes. The dataset includes 14 attack types grouped into five categories: DoS/DDoS (TCP SYN flood, UDP flood, HTTP flood, ICMP flood), information gathering (port scanning, OS fingerprinting, vulnerability scanning), man-in-the-middle (DNS spoofing, ARP spoofing), injection (XSS, SQL injection, uploading), and malware (backdoor, password cracking, ransomware).

The original dataset contains over 20 million records with a heavy class imbalance. For this study, we used the ML subset provided by [18] and subsampled each attack class to a maximum of 750 instances to achieve an approximate 70/30 normal-to-attack balance, resulting in 24,301 normal flows and 10.500 attack flows.

Unlike the Westermo dataset, which was recorded at multiple physical locations in the same network and therefore provides naturally distributed data, most publicly available network traffic datasets are collected as a single centralized file. To enable local and federated experiments on such datasets, synthetic partitioning is required. The partitioning strategy follows the principle of label distribution skew described in [19], where each client is dominated by certain classes while retaining smaller proportions of the remaining classes.

Each attack type was assigned a dominant node: Node 1 received DDoS variants, Node 2 received backdoor, password, ransomware and MITM attacks, and Node 3 received the remaining six attack types. The dominant node received 80% of its assigned attack flows while each of the other two nodes received 10%. Normal traffic was distributed evenly across all three nodes.

B. Preprocessing

Both datasets were preprocessed using a shared pipeline with dataset-specific configuration files, following the procedure described in [14].

For the Westermo dataset, twelve columns were removed: six address-related and six time-related. As noted in [14], including address information would make anomaly detection trivial since most anomalies originate from a single attacker node. The protocol column was one-hot encoded using all six protocol values to ensure consistency across nodes. The 79 Good-SSH flows were relabeled as Normal, and missing values were filled with zero. The NST labeling strategy was selected for both binary and multiclass classification. After preprocessing, each node contained 53 features.

For the Edge-IIoT dataset, 28 columns were removed, including IP addresses, timestamps, payload data and redundant protocol-specific fields. Missing and infinite values were filled with zero. The existing Attack_label and Attack_type columns were used for binary and multiclass classification respectively.

For both datasets, Min-Max normalization to the range [0, 1] was applied after splitting the data into training and test sets. Normalization parameters were computed from the training split only and then applied to the test split, following [14],

to prevent data leakage. Binary features, including one-hot encoded protocol columns, were left unchanged.

C. Experimental Setup

All experiments use 5-fold stratified cross-validation, where each fold uses 80% of the data for training and 20% for testing. Performance is measured using weighted F1-score as the primary metric, with accuracy, precision and recall reported for completeness. Per-attack-type recall is also computed to identify which specific attack types each method detects or misses. Results are reported as the mean across the five folds, consistent with [14].

For RF, hyperparameters are tuned within each fold by further splitting the 80% training data into training and validation. A grid search over the number of estimators (10, 50, 100) and splitting criterion (gini, entropy) selects the combination with the highest weighted F1-score on the validation set. The final model is then retrained on the full training data using the selected parameters. RF is evaluated on both binary anomaly detection (AD) and multiclass anomaly classification (AC).

IF is configured with 100 estimators and a maximum of 256 samples per tree, following [8]. It is trained on normal traffic following the semi-supervised approach [20], reflecting a scenario where historical normal traffic is available but attack samples may not be. Semi-supervised training was adopted after unsupervised training (on all traffic) was found to degrade performance on Edge-IIoT, where attack flows constitute approximately 30% of each node's data. Since IF produces continuous anomaly scores, an adaptive threshold is applied: K-means clustering with two clusters is fitted to the scores, and the threshold is set to the mean of the two cluster centers. K-means yielded the most reliable threshold compared to Gap statistic and Otsu's method. IF is evaluated on AD only; to assess which anomaly types IF detects, the known attack labels of flagged flows are examined.

Three training approaches are compared. In the centralized approach, data from all nodes is merged and a single model is trained and evaluated on the combined dataset. In the local approach, each node trains its own model using only its local data. Each model is evaluated on its own test set and on the test sets of the other nodes individually, to measure cross-node generalization. An additional aggregation (All) combines predictions across all test nodes to enable comparison with the centralized baseline.

In the federated approach, the individual models are combined on a central server without sharing any training data. Local models are first trained on each node as in the local approach. For RF and IF, the global model is created by merging the trees from all local clients through multiple rounds; in each round, the oldest trees in the global model are replaced with new trees from the local clients based on a FIFO data structure. The aggregated global model is then sent back to each node. The resulting global model is evaluated on each node's test set to adhere to federated training practices.

VI. Results

A. Centralized Training

1) Westermo Dataset: The centralized RF model achieved an average accuracy of 0.875, with a weighted F1-score of 0.859 over the five cross-validation folds. Precision reached 0.889 and recall 0.875. The IF model achieved a precision of 0.81, a recall of 0.47 and an F1-score of 0.60. The confusion matrices (Figure 1) illustrate the difference between the models. Random Forest correctly classifies most normal traffic and detects a larger number of anomalies, with relatively few misclassifications. In contrast, Isolation Forest produces more false positives and misclassifies a higher number of anomalies as normal traffic, indicating weaker detection performance. The per-attack-type recall results (Figure 2) reveal differences across attack categories. RF achieves high recall (0.9 or higher) for most attack types, including BAD-MITM, BAD-PORTSCAN1, BAD-PORTSCAN2 and BAD-SSH. Both models show very low recall for BAD-MISCONF and BAD-MISCONF-DUP, with Isolation Forest performing slightly higher but still below 0.25.

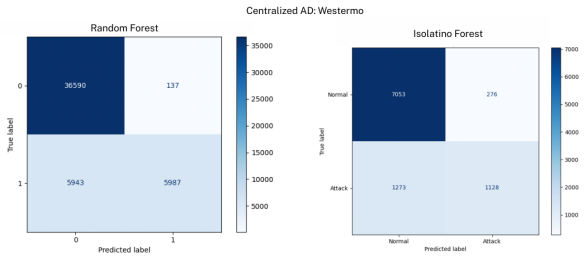


Figure 1. Confusion matrices for centralized anomaly detection on Westermo dataset, summed over 5 folds.

2) Edge-IIoT Dataset: RF achieved an accuracy of 0.954, precision of 0.954, recall of 0.954 and F1-score of 0.954. IF performed substantially worse across all metrics: accuracy 0.472, precision 0.282, recall 0.481 and F1-score 0.355. The confusion matrices (Figure 3) illustrate the gap: RF produced

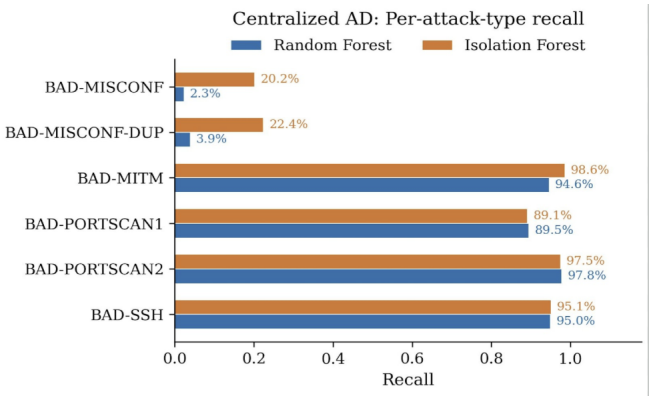


Figure 2. Per-attack-type recall for Random Forest and Isolation Forest centralized anomaly detection on the Westermo dataset, aggregated over 5 folds.

23.618 true negatives and 9.597 true positives with only 683 false positives and 903 false negatives, while IF generated 12,915 false positives and misclassified 5,446 anomalies as normal. Per-attack recall (Figure 4) shows RF detecting all 14 attack types at 77% or above, with perfect recall on DDoS_ICMP, DDoS_TCP, DDoS_UDP and MITM. RF's weakest categories were application-layer attacks (77–81% recall). IF reached 100% only on DDoS_ICMP, DDoS_UDP and MITM, and fell below 32% on eight of the fourteen types.

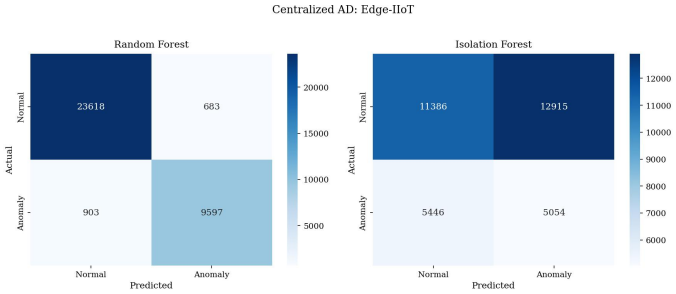


Figure 3. Confusion matrices for centralized anomaly detection on the Edge-IIoT dataset, summed over 5 folds.

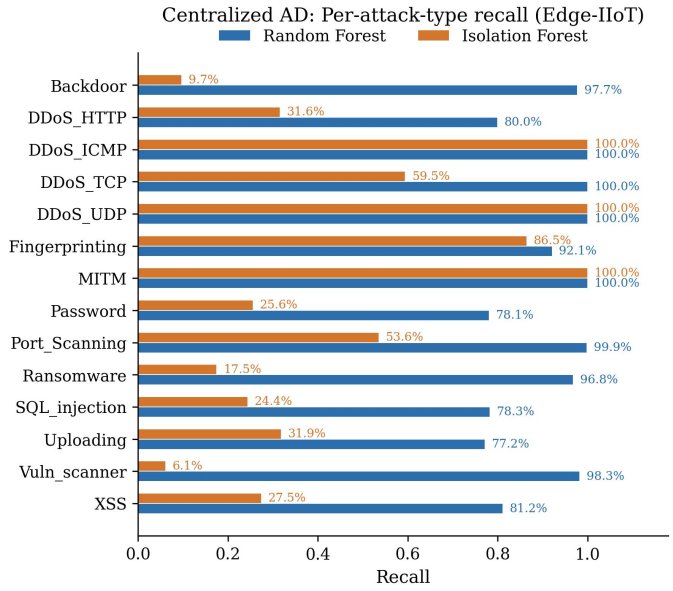


Figure 4. Per-attack-type recall for Random Forest and Isolation Forest centralized anomaly detection on the Edge-IIoT dataset, aggregated over 5 folds.

B. Local Training

Local training produced different generalization patterns depending on the dataset and model. On Westermo, cross-node performance was asymmetric for both models: RF F1-scores ranged from 0.386 to 0.895 and IF from 0.295 to 0.870, depending on the training and test node combination (Figure 5). The Bottom-trained model was the best generalizer for both RF and IF, achieving the highest overall scores (RF All: 0.848 vs. centralized 0.859; IF All: 0.499 vs. centralized 0.60). The Left node reached the highest self-test F1 for both models (RF: 0.895, IF: 0.870) but generalized poorly to other nodes. On Edge-IIoT, RF showed minimal cross-node variation (F1: 0.938 to 0.967, spread of 0.029), with overall scores matching the centralized baseline (0.950 to 0.956 vs. 0.954) (Figure 6, left). IF performed poorly regardless of training node (F1: 0.362 to 0.371, spread of 0.009), confirming a systematic limitation of the unsupervised approach on this dataset (Figure 6, right).

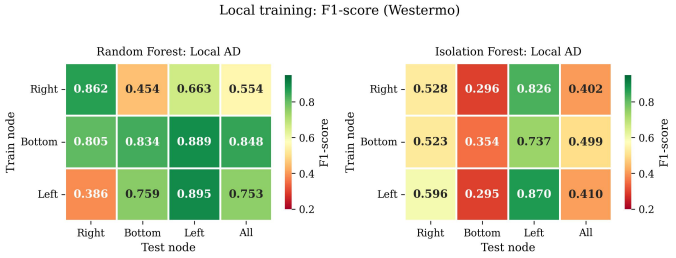


Figure 5. Cross-node F1-score matrices for local training on the Westermo dataset. Each cell shows the F1-score when a model is trained on the row node and tested on the column node. The All column shows performance on the combined test set.

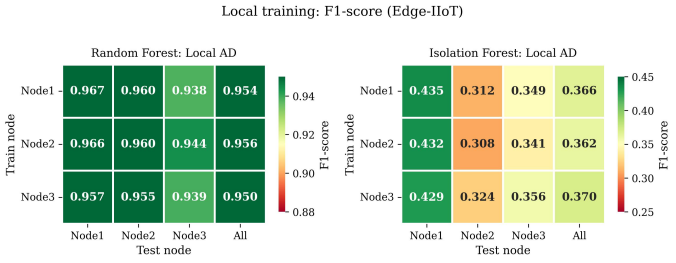


Figure 6. Cross-node F1-score matrices for local training on the Edge-IIoT dataset. Each cell shows the F1-score when a model is trained on the row node and tested on the column node. The All column shows performance on the combined test set.

C. Federated Learning

Federated training on the Westermo dataset showed weak performance in binary classification (AD) for both models; IF achieved a weighted F1-score of 66.5% and RF reached 32%. Multiclass classification (AC) yielded some improvement for RF at 80.04% (Figure 7). The Edge-IIoT dataset showed similar results for IF with a weighted F1-score of 55.6%, while RF reached 98.72% (AD) and 94.3% (AC) (Figure 8).

VII. Discussion

This section discusses the results in relation to the research questions.

A. Performance across training approaches

Comparing the three training strategies reveals a clear pattern: centralized training provides the strongest baseline, local training can approach it under the right conditions, and federated training introduces challenges that reduce effectiveness.

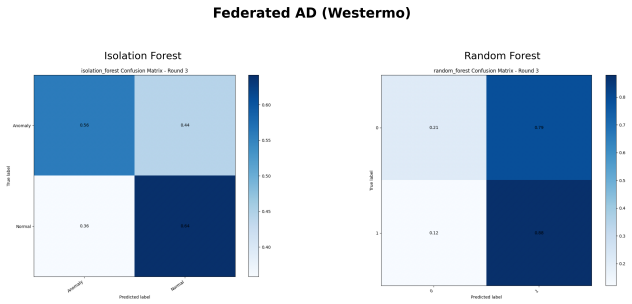


Figure 7. Confusion matrices for federated training binary classification showing predicted/actual label for the last round on Westermo dataset.

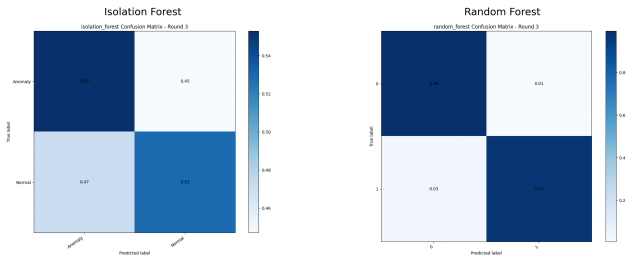


Figure 8. Confusion matrices for federated training binary classification showing predicted/actual label for the last round on Edge-IIoT dataset.

For RF, centralized training achieved F1-scores of 0.859 on Westermo and 0.954 on Edge-IIoT. Local training nearly matched these when the training node had a representative traffic distribution; the Bottom-trained RF model on Westermo reached an overall F1 of 0.848 compared to the centralized 0.859, and local RF models on Edge-IIoT scored between 0.950 and 0.956. The strong local performance on Edge-IIoT reflects the synthetic 80/10/10 partitioning, which gives each node enough spillover exposure to learn all attack signatures. On Westermo, natural node separation creates non-IID distributions with different traffic characteristics per node, resulting in asymmetric generalization gaps.

For IF, the pattern is consistent but at a lower performance level. Centralized IF achieved F1-scores of 0.60 on Westermo and 0.355 on Edge-IIoT, and local IF showed similar numbers regardless of which node provided the training data (F1 approximately 0.36 on Edge-IIoT). This stability suggests that IF's limitation is inherent to the unsupervised method rather than a consequence of data distribution.

Federated training produced the weakest results, particularly for RF on Westermo (F1 32.08% for AD). This stems from a combination of factors: non-IID data distributions across nodes, limitations in the aggregation methods, lack of hyperparameter tuning for the federated setup, and the strict constraint of no raw data sharing between clients. For RF, increasing the number of decision trees did stabilize performance, but also increased the model size transmitted per round. Increasing the number of trees did not appear to affect IF.

On Edge-IIoT, the pattern differed: federated RF achieved 98.72% (AD) and 94.3% (AC). The synthetic partitioning ensures that class diversity remains high across clients, reducing the dominance of any single client's bias. The multiclass results on Westermo support this; federated RF reached 80.04% F1 in AC, substantially better than 32.08% in AD, suggesting that richer class information helps the aggregated model retain discriminative capacity. The virtual federated setup does not reproduce the constraints of a physical deployment, and results from a real environment may differ.

B. Detection and Precision-recall trade-offs

The two models exhibit different detection behavior. RF produces balanced precision and recall across both datasets. IF faces a precision-recall dilemma stemming from its threshold-based separation of normal traffic and anomalies.

On Westermo, both models show very low recall for BAD-MISCONF and BAD-MISCONF-DUP (below 0.25). These misconfiguration-type anomalies produce traffic patterns that closely resemble normal traffic, making them difficult to isolate for both methods. For all other attack types on Westermo, RF maintains recall above 0.9, while IF shows more variable performance. On Edge-IIoT, the contrast becomes sharper. RF detects all 14 attack types at 77% recall or above, with perfect recall on the volumetric DDoS variants (ICMP, TCP, UDP) and MITM. IF also reaches 100% on these same volumetric types but drops below 32.08% on eight of the fourteen categories. Particularly notable is IF's poor detection of Vulnerability_scanner (6.1%), Backdoor (9.7%) and Ransomware (17.5%), whereas RF catches these at over 96% recall. These attack types appear to lack the distinct feature-space separation that IF depends on.

The threshold challenge is central to IF's limitations. The contamination factor cannot be determined autonomously by the model; we addressed this with K-means clustering, aligning with [20]. When anomalies overlap substantially with normal traffic, K-means cannot establish a reliable boundary. Lowering the threshold increases false alarms; raising it causes subtle attacks to be missed. IF's 12,915 false positives on Edge-IIoT, nearly 19 times RF's 683, illustrate this and would make IF impractical where alert fatigue is a concern.

IF is generally more sensitive to noisy and irrelevant features than RF [8], which motivated the removal of features such as unique IDs and timestamps during preprocessing. Even after this step, IF underperformed RF in all configurations. IF is also susceptible to swamping, where normal points are flagged as anomalies, and masking, where groups of anomalies form clusters classified as normal [9].

These detection patterns remained consistent across training approaches. IF's limitations on Edge-IIoT persisted regardless of training strategy (F1 between 0.355 and 0.371), reinforcing that the weakness is inherent to the method rather than of data partitioning.

C. Generalisation across datasets

The core findings hold across both datasets, but the magnitude of the differences shifts. The F1 gap between RF and IF on

Edge-IIoT (0.954 vs. 0.355) is larger than on Westermo (0.859 vs. 0.60), indicating that the supervised advantage grows with attack diversity. Edge-IIoT contains 14 attack types spanning five categories compared to Westermo's six, and the greater variety amplifies the challenges IF faces.

The relative performance of the three training approaches follows the same pattern on both datasets. Federated training shows the greatest variation: RF drops on Westermo (32.08% AD) but performs well on Edge-IIoT (98.72% AD), reflecting the difference between natural non-IID partitioning and synthetic splits that retain minority class representation.

The per-attack detection patterns also generalize. On both datasets, IF performs well on volumetric attacks with clear feature-space outliers (port scans, DDoS floods, MITM) but struggles with subtle attack types. RF detects both categories reliably, suggesting that the relative strengths of the two models reflect fundamental differences in detection approach rather than characteristics of a particular dataset.

Two limitations apply to the generalizability assessment. The Edge-IIoT partitioning is synthetic and does not reflect the natural variation in data distributions found in real deployments; Westermo's natural node distributions provide a more realistic test of federated and local training. The datasets also differ in feature extraction methodology, number of features and traffic composition, making direct numerical comparison less meaningful than comparing patterns and trends.

VIII. Conclusion

This study compared Random Forest and Isolation Forest for anomaly detection in industrial network traffic under centralized, local and federated training, evaluated on two datasets with different network characteristics and attack compositions. Random Forest consistently outperformed Isolation Forest across all training approaches and both datasets. Centralized and local training produced the strongest results, while federated training underperformed primarily due to non-IID data distributions and aggregation limitations. The supervised advantage grew with attack diversity, and the findings generalized across both datasets. A standalone Isolation Forest is not optimal for real-time industrial anomaly detection, but combined with a secondary model that addresses the threshold problem, it could contribute to a hybrid system capable of identifying threats that supervised models alone might overlook.

Several areas remain unexplored. Hyperparameter tuning was not performed in the federated setting, which likely limited its performance. All experiments were conducted in a virtual simulation that does not capture real-world constraints such as bandwidth limitations and hardware differences between nodes. Deploying federated models directly in network routers and combining rule-based detection with machine learning to improve detection of misconfiguration attacks are directions that could address limitations observed in this study.

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